Industrial Powder Coat Rack

Summary: Designed and deployed a rack assembly for an industrial powder-coating line to improve loading time, space efficiency, and system mobility.

Tools: Inventor, AutoCAD, FEA, Excel

Key Constraints: 500°F oven environment, **~800 lbf** loading (factor of safety **1.5**), compatibility with existing conveyor system

Process:

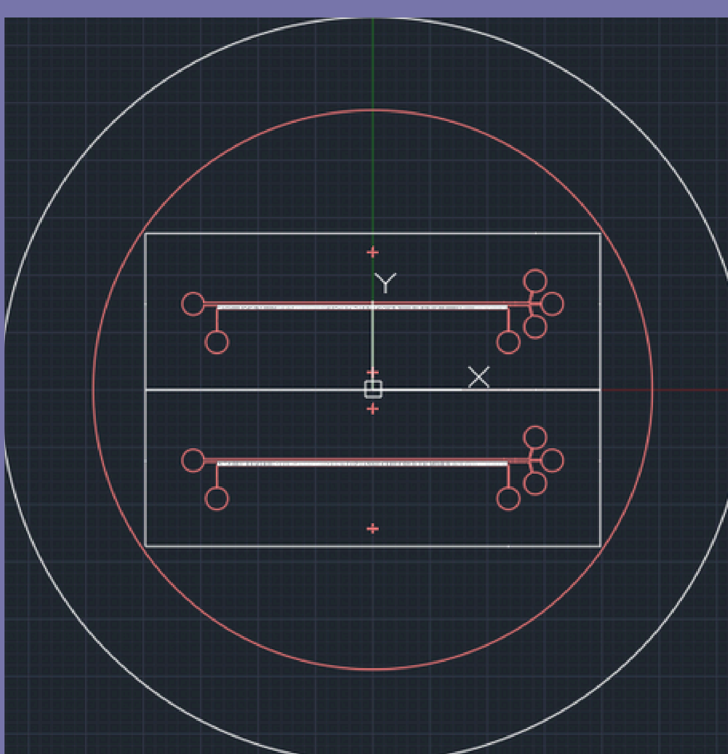
- Built full 3D part and assembly models in Inventor
- Validated structural integrity using FEA, ensuring a **1.5×** factor of safety
- Generated AutoCAD production drawings to company standards, feeding directly into laser cutting and fabrication workflows
- Coordinated in-house manufacturing and part sourcing from external vendors while maintaining cost estimates within budget
- Deployed rack cut part loading time by **~60%** and improved space utilization and heat uniformity

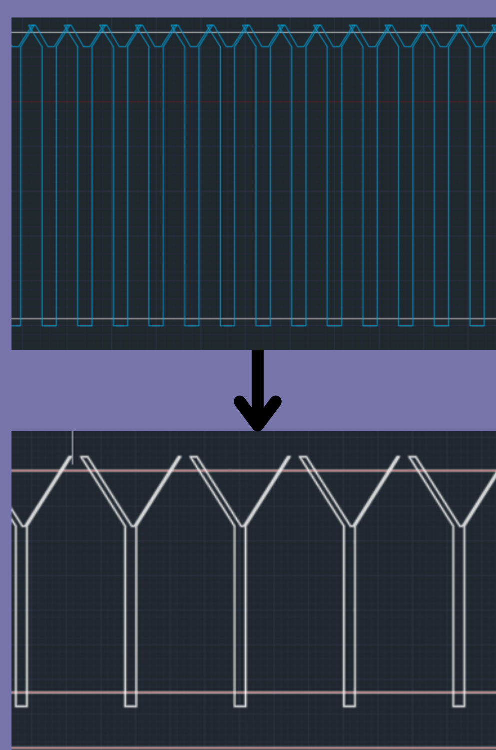
Fully assembled rack, two views

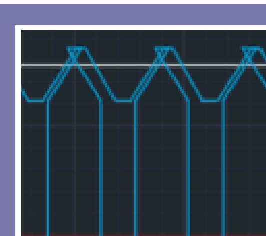
MAqCI Design

Summary: Designed a multi-layer photomask design for MAqCI device in AutoCAD, resolving micron-scale channel-spacing and functional constraints through iterative design to produce a manufacturable design.
Goal: Create a chip that can extract a large amount of information from a small biomass (ex: a drop of blood)
Process:

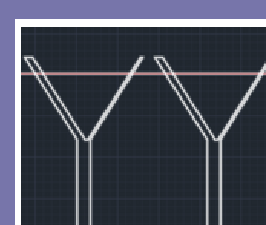
- Developed initial design consisting of 200+ 400-µm-long and 10-µm-tall Y-channels at a 65-degree bifurcation angle based on publication
- Review showed incompatibility between published channel spacing and fabrication constraints
- Iteratively modified design geometry to achieve manufacturable layout
- Finished design consists of two mask layers to account for two different feature heights of 10 µm and 50 µm features, as seen below.
- Y-channels slightly overlap with medium and cell-seeding channels to account for alignment error during photolithography







Initial y-channel design

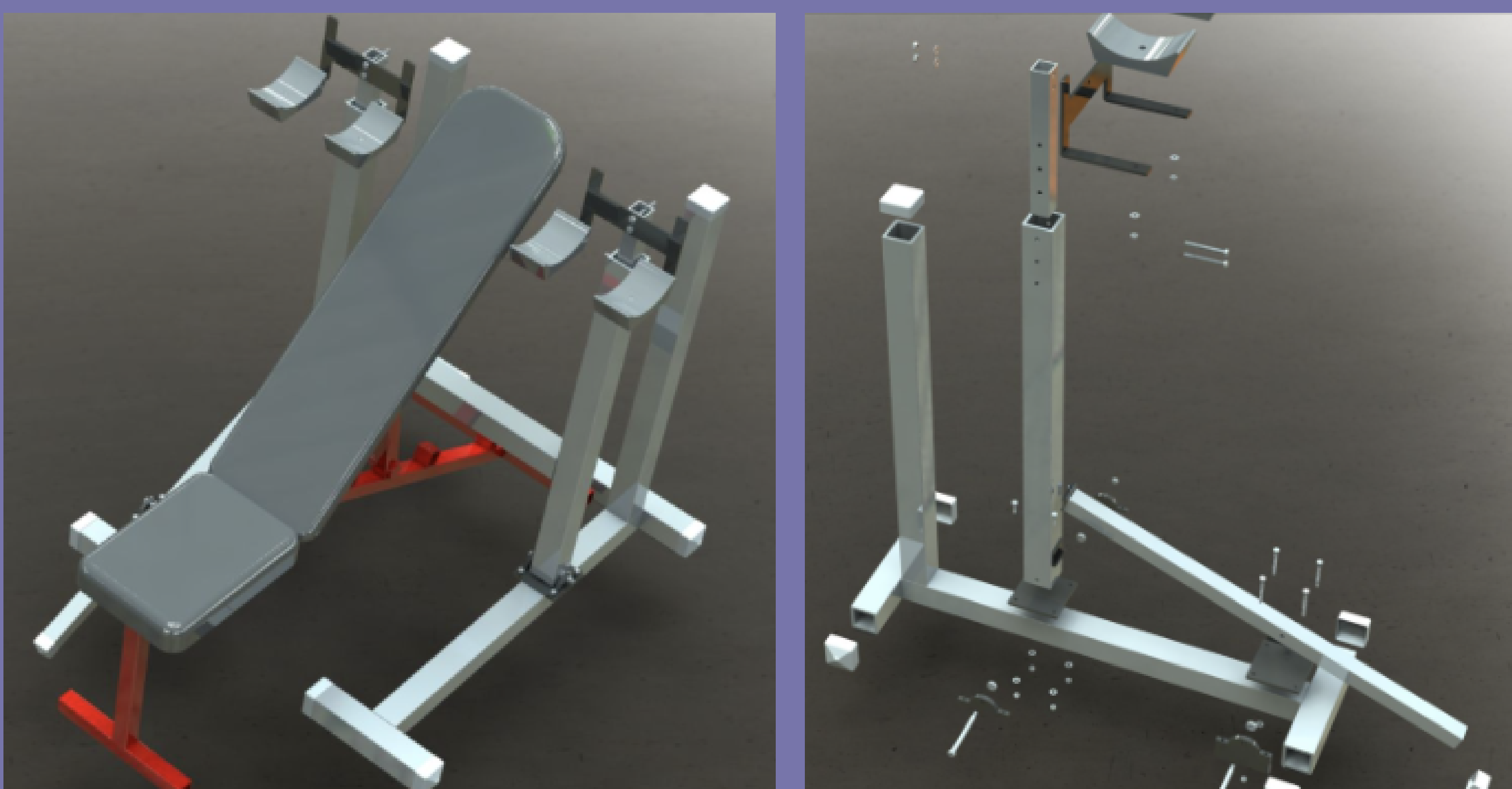


Final y-channel design

Dumbbell Press Machine

Summary: Designed a complete system to support the dumbbell press exercise, progressing from concept development to a product proposal presentation with manufacturing considerations and photorealistic visualizations.
Tools: Solidworks
Process:

- Developed initial concept sketches and iterated through multiple team-based design revisions to refine functionality and ergonomics
- Performed engineering calculations to size structural members, select materials, and identify required off-the-shelf components
- Modeled all custom parts in SolidWorks and produced fully-dimensioned manufacturing drawings
- Created finalized design report including an exploded view and photorealistic renderings to communicate assembly sequence and design intent
- Presented the finalized design to a class of 70 engineering students



Proof of Concept (left), Exploded view (right)